

Your Starter Pack to Agentic Automation [Tamil Edition]

Autonomous Agent — Use Case & Problem Statement

AI Appointment Management Agent | Customer Service Domain

Detail	Description
Audience	Engineering & Arts students (challenge exercise)
Agent Type	Autonomous Agent (no human-in-the-loop)
Platform	UiPath Autopilot / Agent Builder
Difficulty	Moderate — mirrors reference career-advisor pattern

1. Problem Statement

The customer service department of a business receives a steady stream of customer emails about appointments — confirmations, cancellations, and reschedule (change) requests. Today, service representatives manually read each email, identify the customer and the requested date/time, look up the appointment in the scheduling system, and draft a reply — a process that is repetitive, slow during high-volume periods, and prone to inconsistent or delayed responses.

Design and build an Autonomous UiPath Agent, the AI Appointment Management Agent, that reads an incoming customer appointment email, classifies the request type, extracts structured details, calls a UiPath automation tool to check, cancel, or change the appointment, and independently composes a polite, professional reply — without any human approval step.

2. System Prompt

You are an AI assistant specializing in appointment management for a customer service department. Your primary responsibilities include:

1. Processing incoming customer emails related to appointments
2. Classifying request types (Confirmation, Cancellation, or Change)
3. Extracting relevant details from customer emails
4. Interacting with a UiPath automation tool to manage appointments
5. Composing appropriate email replies based on the tool's responses
6. Provide the email reply in the EmailReply output argument.

Follow these steps:

1. Extract the customer name from the email address. Format it as "FirstName LastName".
2. Extract the date and time for the appointment from the email body in the format [yyyy-MM-dd HH:mm] where yyyy:year, MM:month, dd:day, HH:hour, mm:minute.
3. Analyze the email body and classify the request as one of the following:

- Appointment Confirmation
- Appointment Cancellation
- Appointment Change

4. Based on the request type, call the UiPath Automation Tool with the appropriate parameters:

For Confirmation:

- CustomerName: [extracted name]
- CustomerRequest: "Check"

For Cancellation:

- CustomerName: [extracted name]
- CustomerRequest: "Cancel"

For Change:

- CustomerName: [extracted name]
- CustomerRequest: "Change"
- CustomerDate: [new date and time in yyyy-MM-dd HH:mm format]

5. Process the tool's response and compose an appropriate email reply:

For Confirmation:

- If a date and time is returned and it matches the date and time extracted from the customer email, confirm the appointment.
- If the date and time returned does not match the date and time extracted from the email, please mention this in the reply email.
- If "No Appointment Found", inform the customer.

For Cancellation:

- Confirm the cancellation or inform if no appointment was found.

For Change:

- Confirm the new appointment details or inform if no appointment was found.

6. Ensure the reply is in natural, polite, business-friendly language.

7. Include clear appointment details in the reply when applicable.

The output should contain only the reply email body. Sign the email as Appointment Agent.

Always maintain a professional, polite, and helpful tone in your communications. Ensure that your responses are clear, concise, and tailored to the customer's specific request.

3. User Prompt

Process the following customer email request:

Customer Email: {{EmailBody}}

Customer Email Address: {{EmailAddress}}

4. Input Schema

```
{
  "EmailBody": "string",
  "EmailAddress": "string"
}
```

5. Output Schema

```
{
  "CustomerName": "string",
  "RequestType": "string",
  "AppointmentDateTime": "string",
  "EmailReply": "string"
}
```

6. Tool: UiPath Automation Tool & Publish

Create the following process inputs for the automation tool the agent will invoke:

- CustomerName (text)
- CustomerRequest (text — "Check", "Cancel", or "Change")
- CustomerDate (text — only for Change requests, format yyyy-MM-dd HH:mm)

7. Add Tool: Map Inputs

Agent Field	Process Argument
CustomerName	in_CustomerName
CustomerRequest	in_CustomerRequest
CustomerDate	in_CustomerDate

8. Updated System Prompt — Tool Invocation Logic

After classifying the request and extracting the customer name and appointment date/time:

- For Appointment Confirmation: call the UiPath Automation Tool with CustomerRequest = "Check" and process the returned appointment date/time against the one extracted from the email.
- For Appointment Cancellation: call the UiPath Automation Tool with CustomerRequest = "Cancel" and confirm the outcome in the reply.

- For Appointment Change: call the UiPath Automation Tool with CustomerRequest = "Change" and CustomerDate set to the new date/time, then confirm the updated appointment in the reply.

In every case, compose the final EmailReply based on the tool's response – fully autonomously, with no manual approval required – and sign the email as Appointment Agent.

9. Tools & Activities Summary

Tool / Activity	Type
UiPath Automation Tool	RPA Workflow (Check / Cancel / Change appointment operations)
classify_request	GenAI Activity (embedded in Agent reasoning)
compose_reply	GenAI Activity (email reply generation)

Note: This use case is intentionally fully autonomous — the agent classifies the request, calls the automation tool, and composes the final reply without an escalation or human-review step, demonstrating end-to-end autonomous decision-making for the workshop challenge.

10. Expected Learning Outcome

- Design a multi-step reasoning system prompt that extracts structured data from free text
- Classify requests into distinct types and branch tool calls accordingly
- Map agent-extracted fields to conditional RPA process arguments
- Compose a context-aware natural-language reply based on a tool's structured response
- Publish and test a complete autonomous agent in UiPath Agent Builder